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Subject : Advanced Cloud Computing (P)

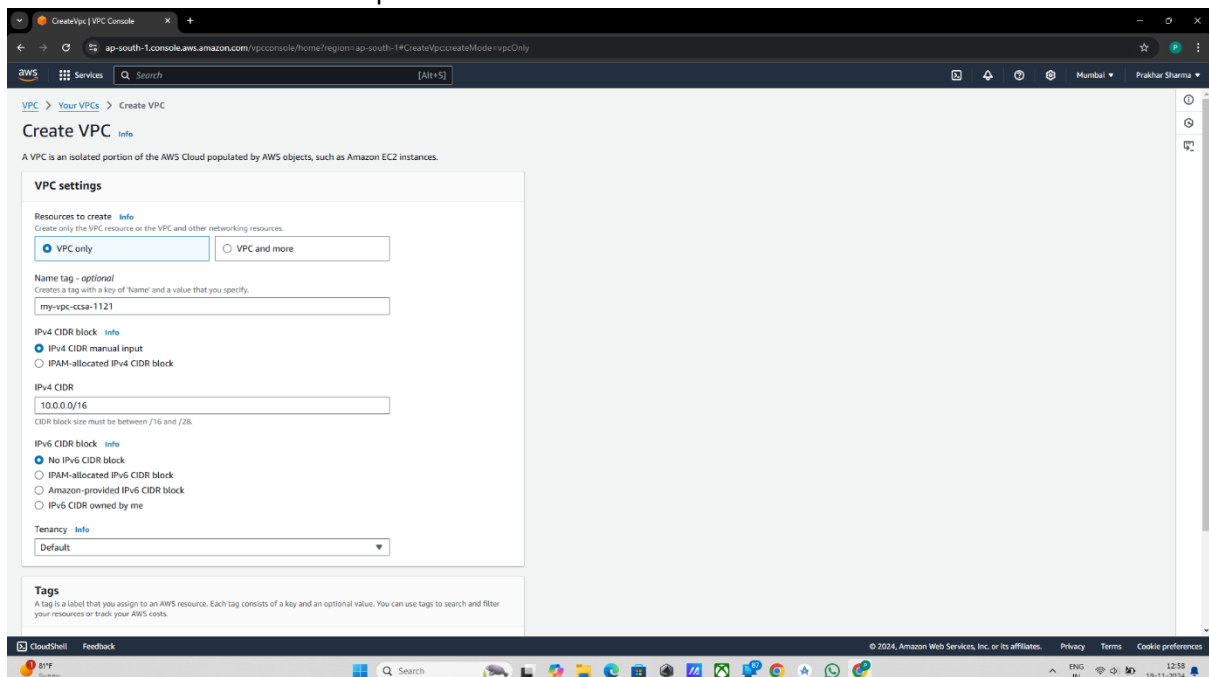
Name of the Student: **Prakhar Anil Sharma**

PRN: 20220801121

Title of Practical: **Configuring Layered Security in an AWS VPC(NACL)**

Step – 1 : Creating VPC .

- 1- Go to Amazon Management Console.
- 2 – Search For VPC Service.
- 3 – Click on Your VPCs in the left panel.
- 4 – Click on create VPC.
- 5 – In VPC Settings , Select VPC only in Resource to create.
- 6 – Name Your VPC.
- 7 – Select IPv4 CIDR manual input option .
- 8 – Give 10.0.0.0/16 as IPv4 CIDR .
- 9 – Select No IPv4 CIDR block option .



- 10 – Click on Create VPC .

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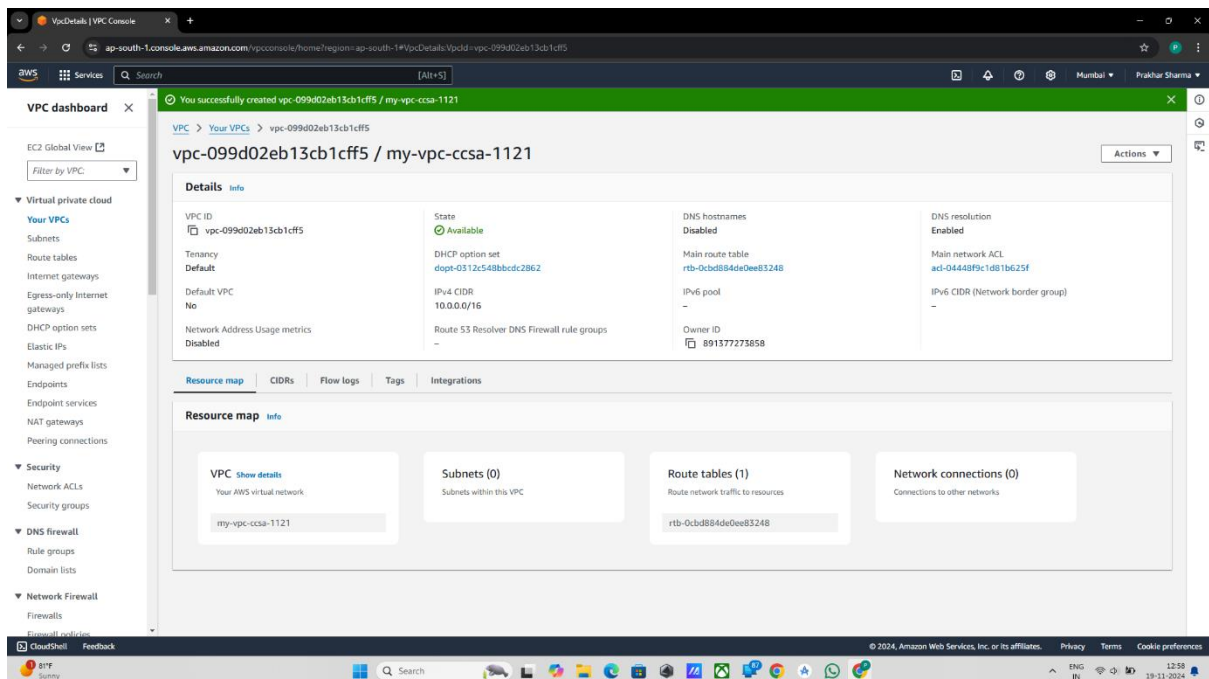
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- 11 – Go to subnets in the left hand panel , click on it.
- 12 – Click on create Subnet .
- 13 – In VPC ID , select the VPC created above from dropdown .
- 14 – In Subnet settings , Name the subnet .
- 15 – Select Availability Zone as per choice .
- 16 – Select IPv4 VPC CIDR block as 10.0.0.0/16 .
- 17 – Set IPv4 subnet CIDR block as 10.0.0.0/24 .
- 18 – Click on create subnet .

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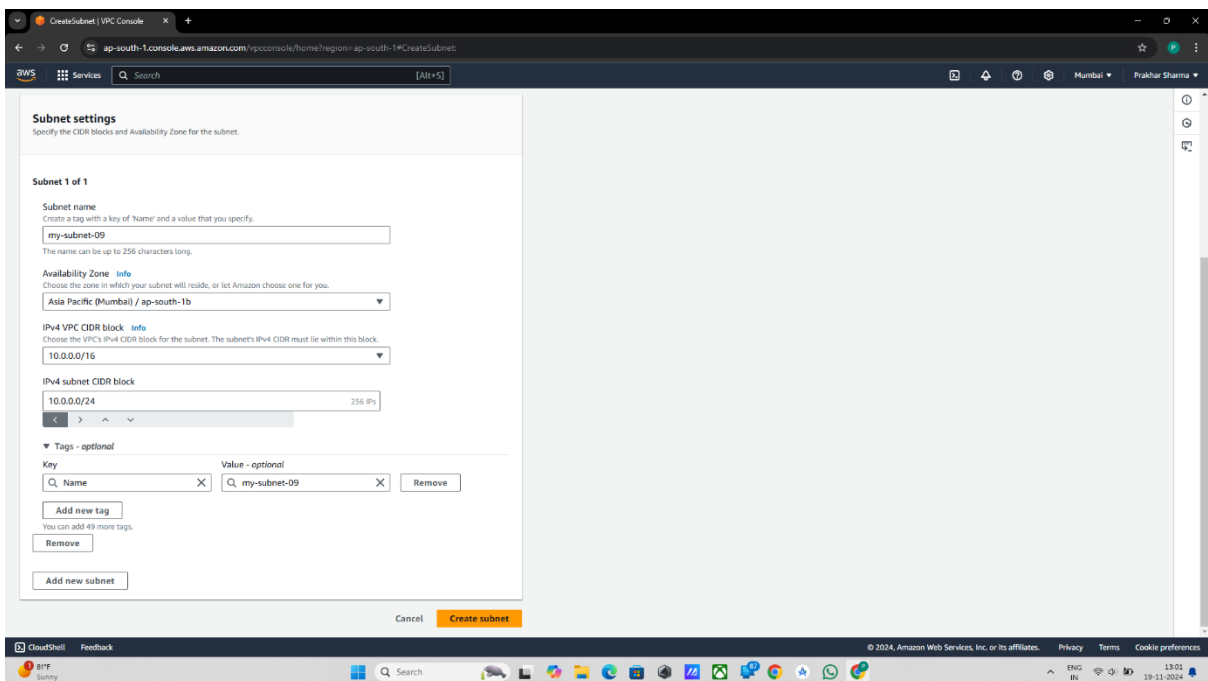
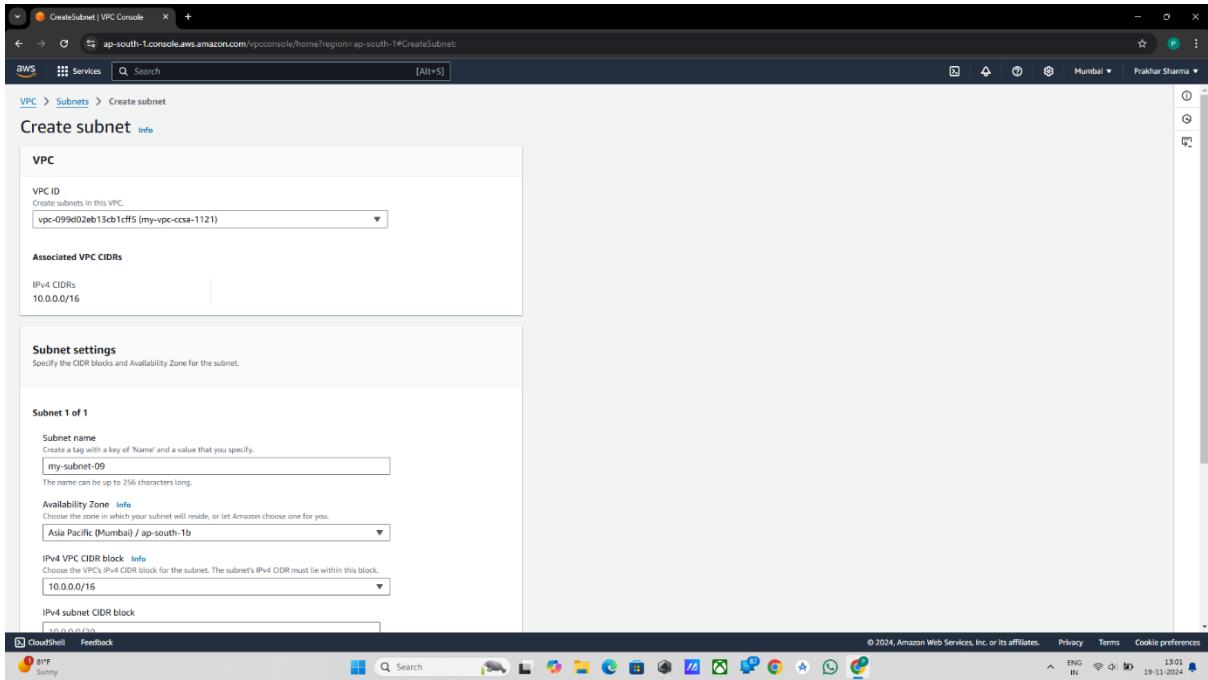
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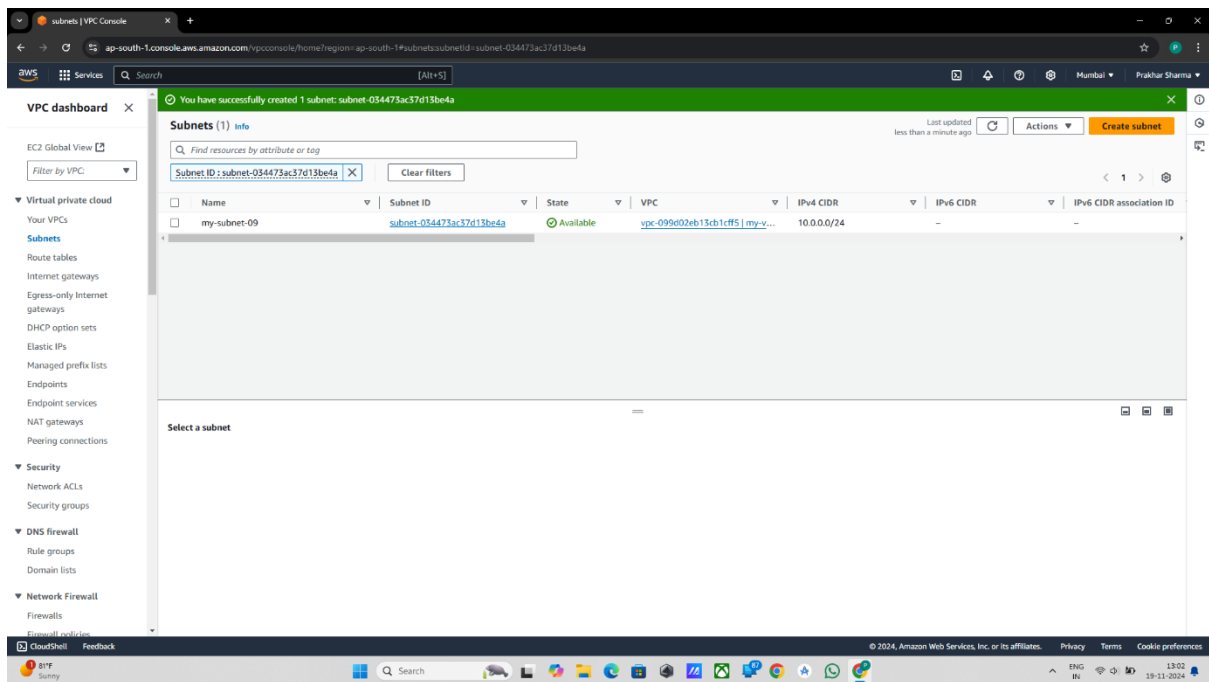
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19 – Go to Internet Gateway in the left panel and click on it .

20 – Click on create internet gateway .

21 – Name the internet gateway .

22 – Click on create in internet gateway .

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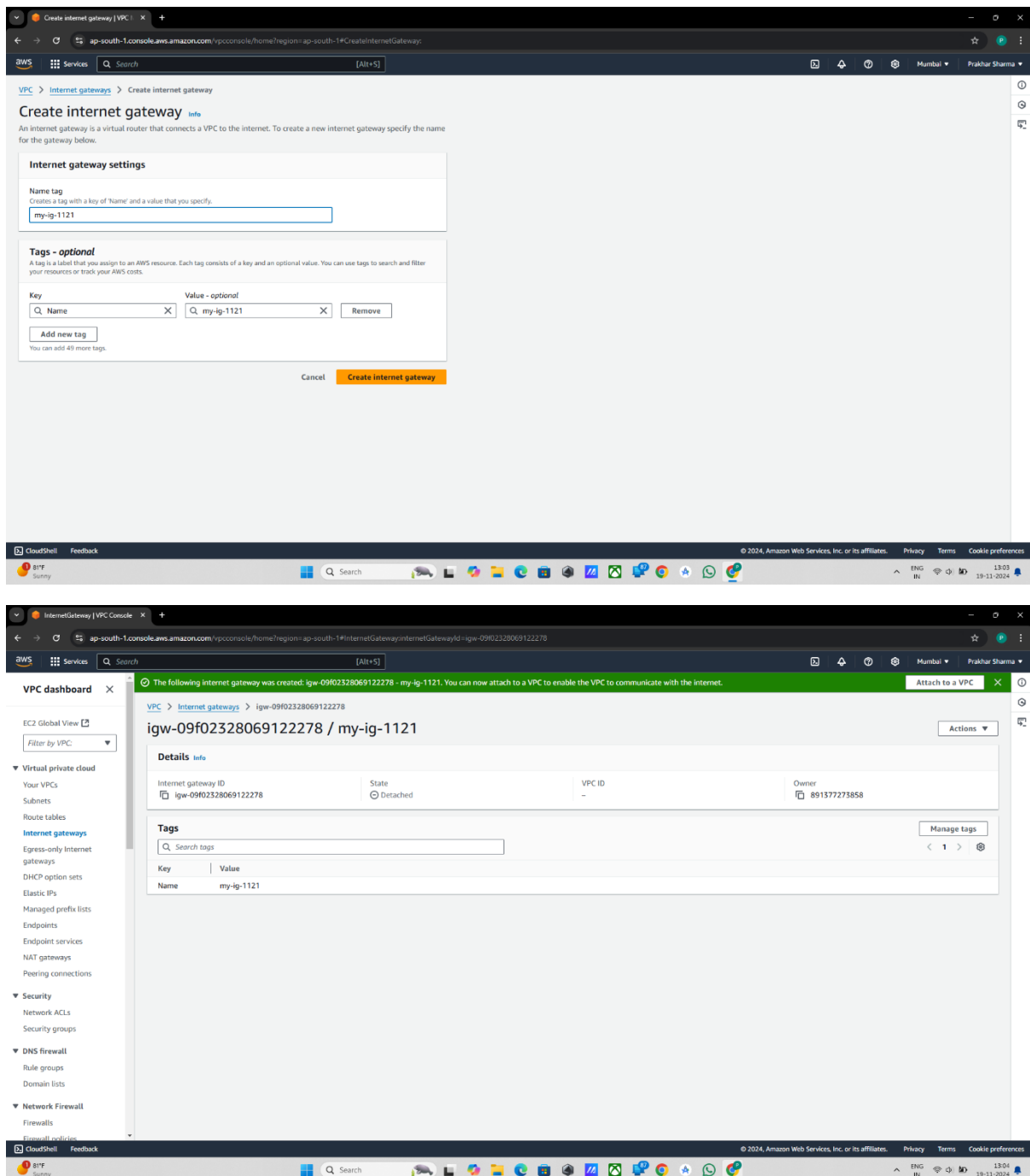
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The screenshot displays the AWS Management Console interface. The top section shows the 'Create internet gateway' wizard, where the 'Name tag' is set to 'my-ig-1121'. Below this, the 'Tags - optional' section is visible, showing a table with one tag: 'Name' with value 'my-ig-1121'. The bottom section shows the 'Internet gateway' details page for the newly created gateway 'igw-09f02328069122278 / my-ig-1121'. The details include the gateway ID, state (Detached), VPC ID, and owner. The 'Tags' section shows the same tag as above.

23 – Select the internet gateway you created .

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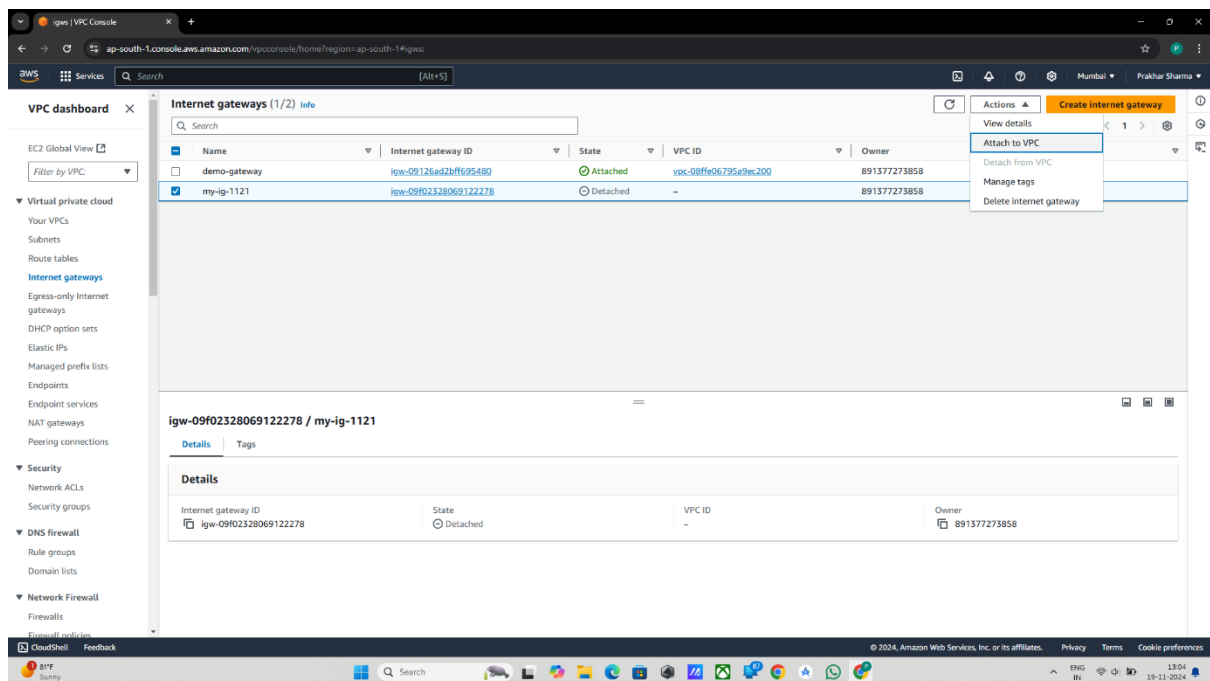
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24 – Go to Action at top right corner and click on it .

25 – Click on Attach to VPC .



26 – Select the VPC you created above .

27 – Click on Attach internet gateway .

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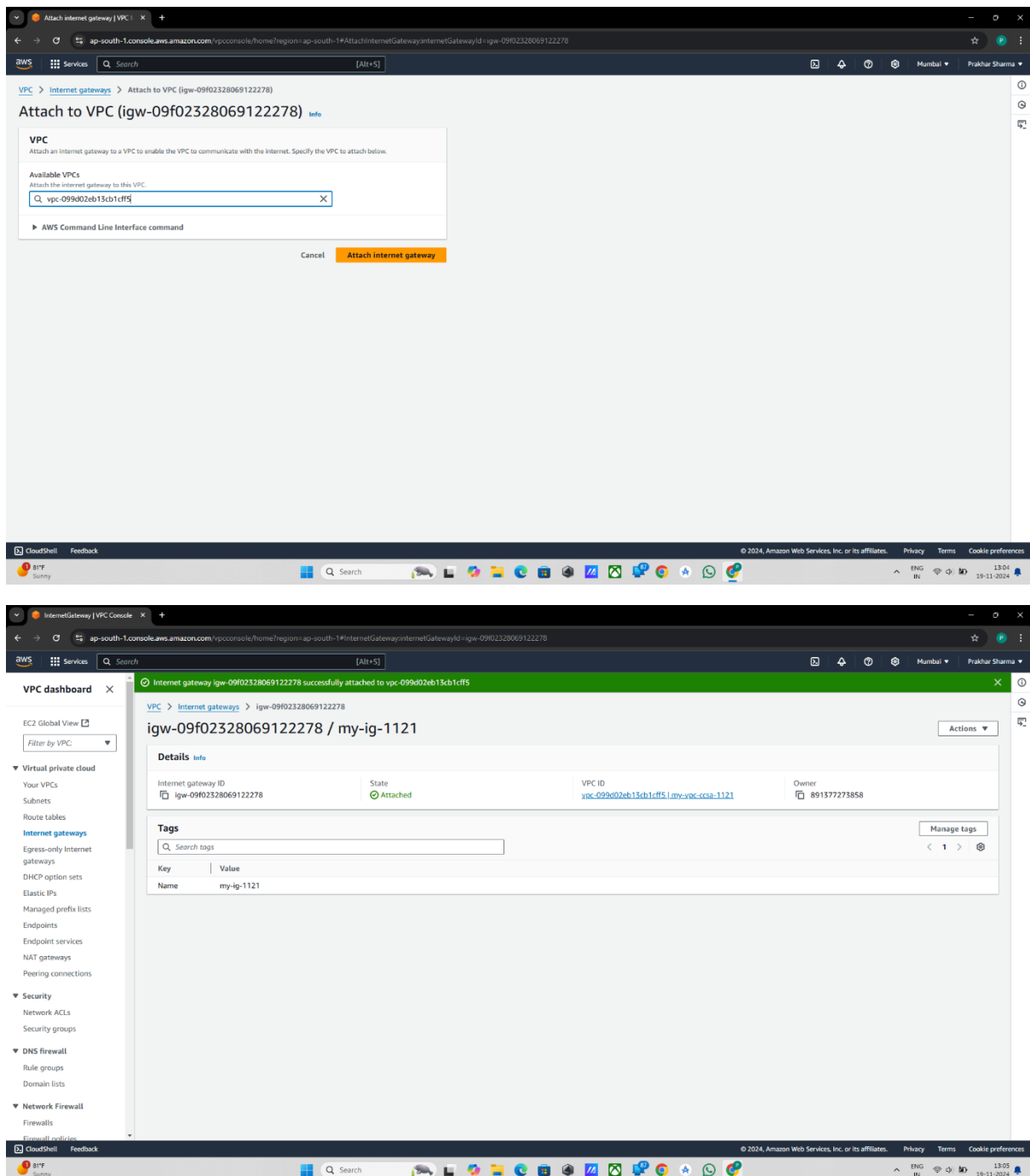
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28 – Go to route table at the left panel , click on it .

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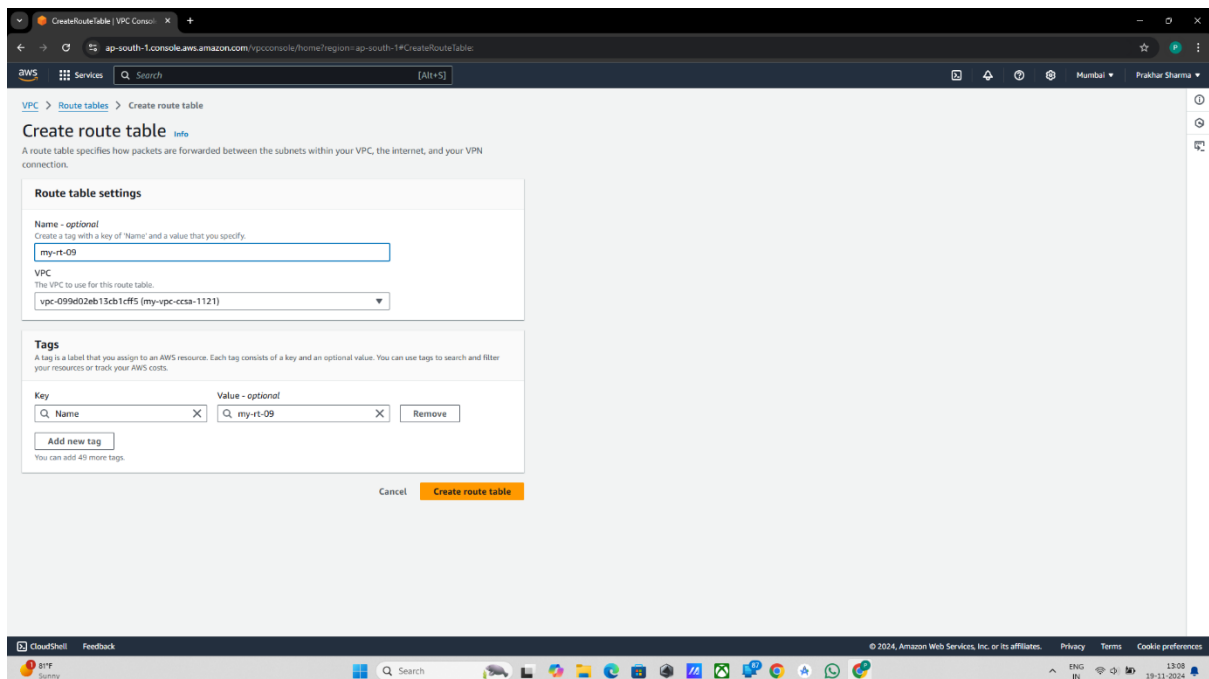
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29 – Name the Route table .

30 – Select the VPC you created from the dropdown .

31 – Click on route table .



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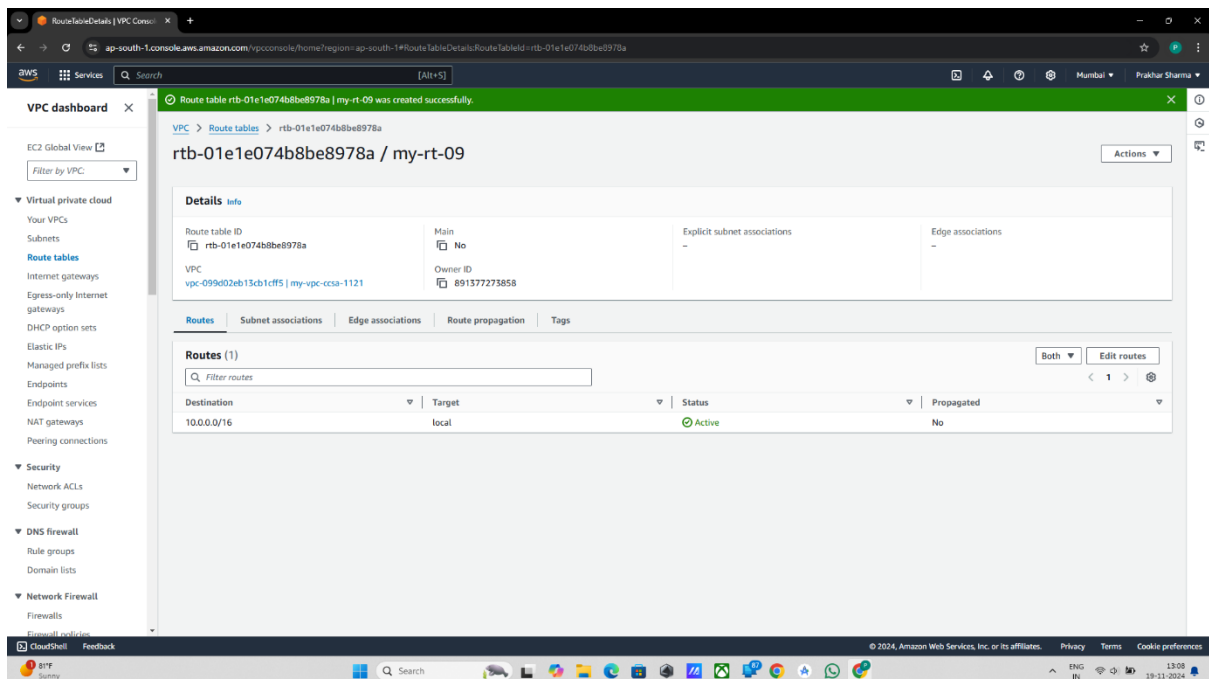
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32 – Click on the route table you created .

33 – At the bottom , Go to Subnet Association .

34 – Click on edit subnet association .

35 – Select the subnet and click on save association .

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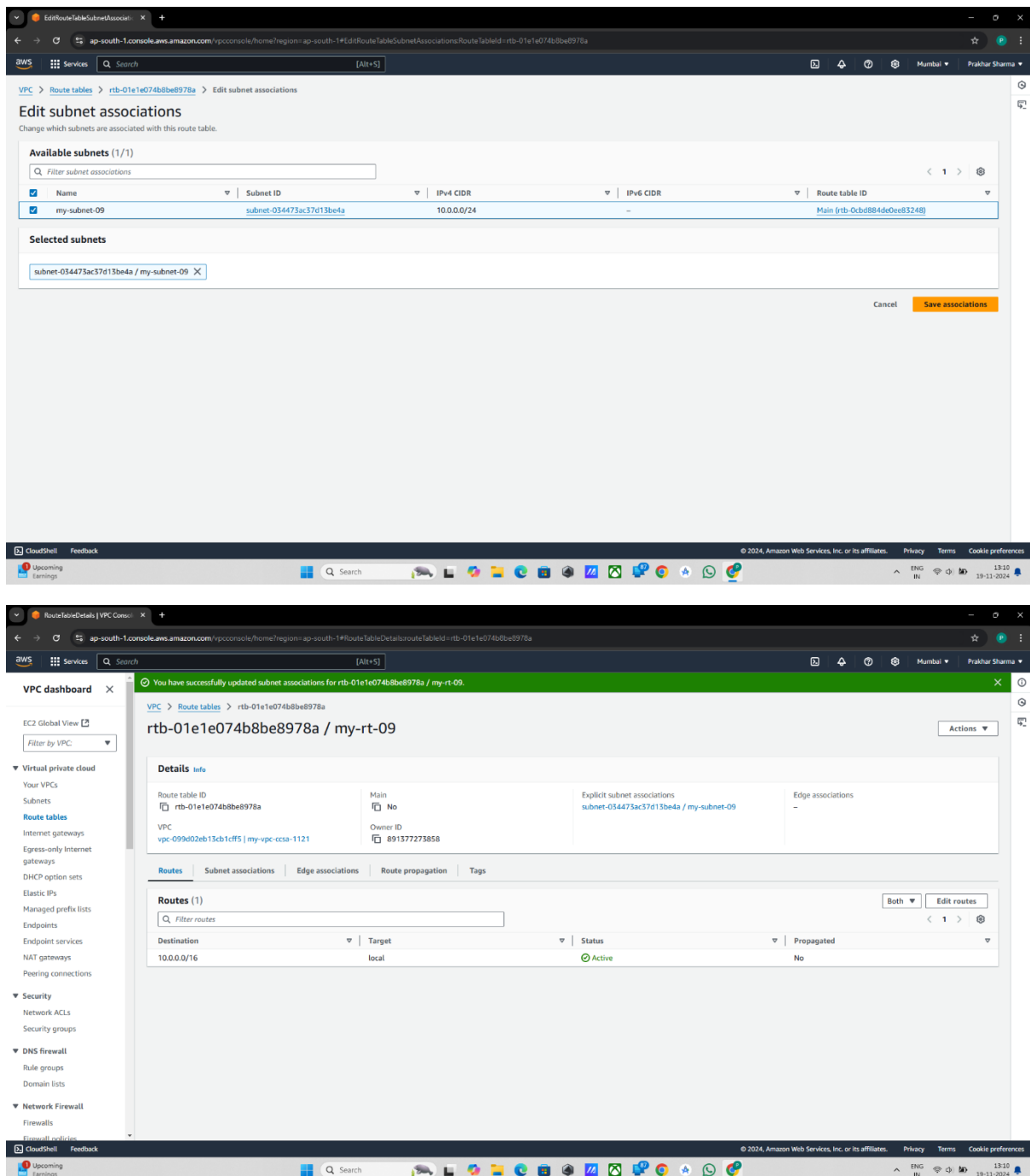
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The screenshot displays the AWS Management Console interface. The top section shows the 'Edit subnet associations' page for route table 'rtb-01e074b8be978a'. It lists available subnets, with 'my-subnet-09' (subnet-034473ac37d13be4a) selected. The bottom section shows the 'Routes' tab for the same route table, displaying a single active route to the local destination 10.0.0.0/16.

Available subnets (1/1)

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
my-subnet-09	subnet-034473ac37d13be4a	10.0.0.0/24	-	Main (rtb-0cb884dc0ee83248)

Selected subnets

subnet-034473ac37d13be4a / my-subnet-09

Details Info

Route table ID	Main	Explicit subnet associations	Edge associations
rtb-01e074b8be978a	No	subnet-034473ac37d13be4a / my-subnet-09	-

Routes (1)

Destination	Target	Status	Propagated
10.0.0.0/16	local	Active	No

36 – Now Go to routes tab at the bottom .

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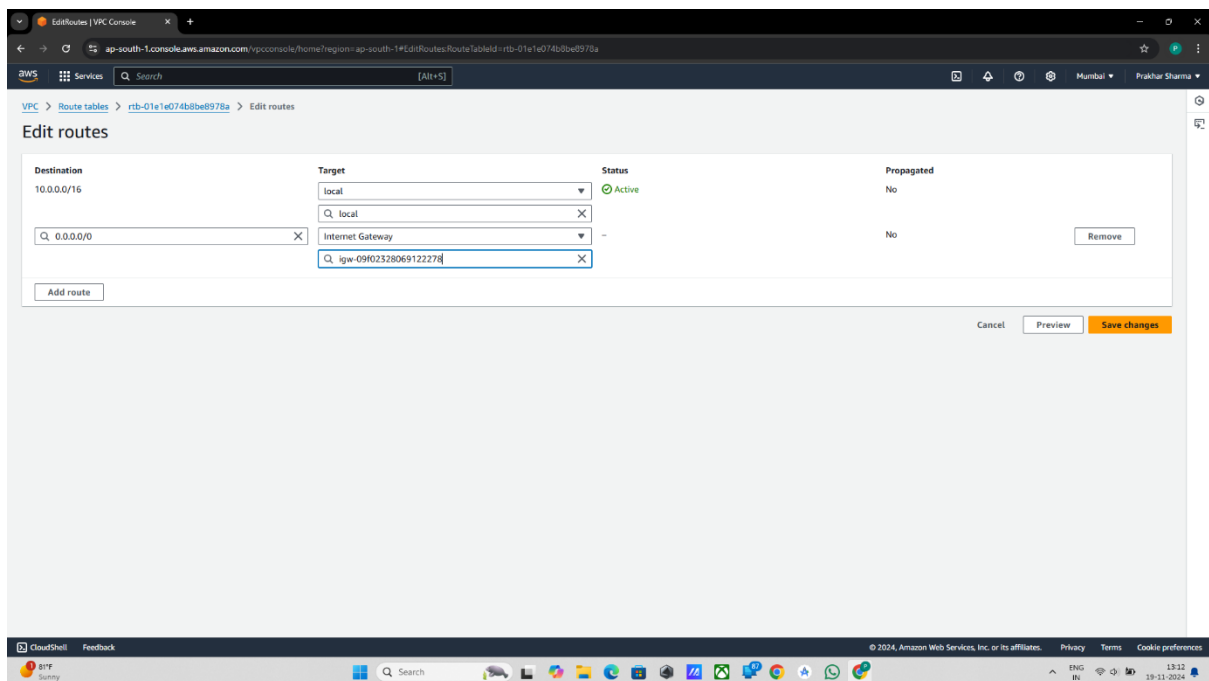
37 – Click on edit route .

38 – Click on add rule .

39 – Select Destination as 0.0.0.0/0 .

40 – In target, Select internet gateway and select the internet gateway created above .

41 – Click on save changes .



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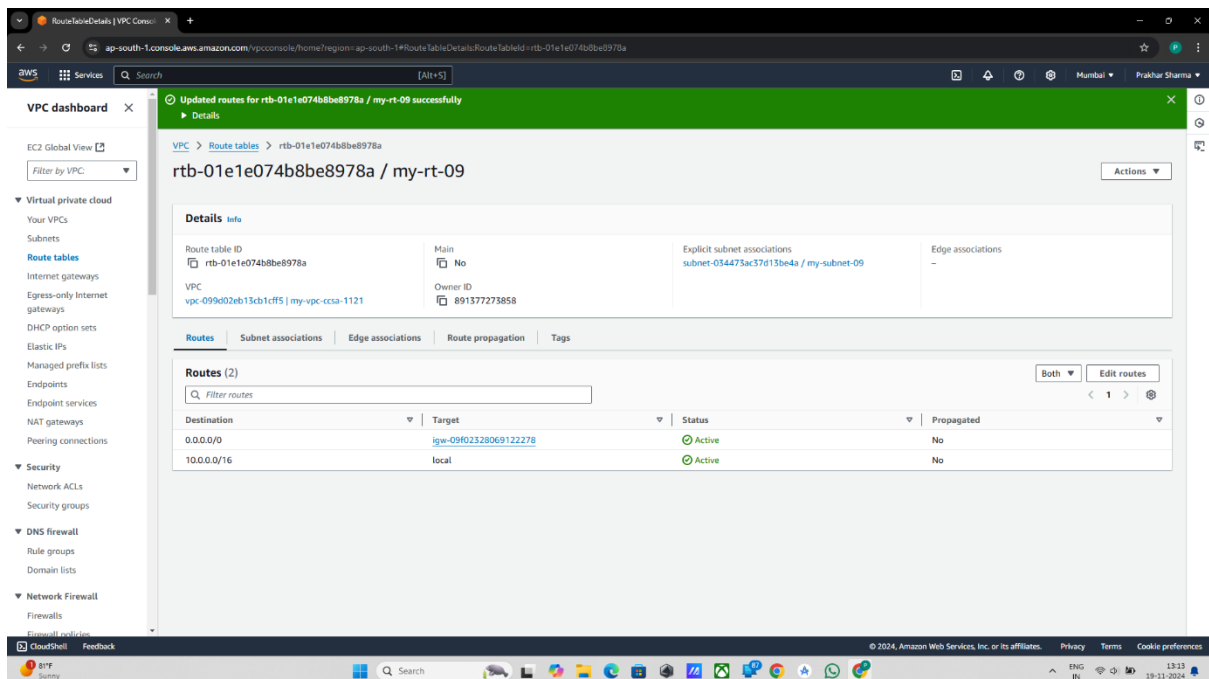
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- 42 – Search for EC2 service .
- 43 – Click on EC2 .
- 44 – Click on Launch Instance .
- 45 – Name the instance .
- 46 – Select AMI as Windows 2022 Base (Free Tier Eligible) .
- 47 – Select instance type as t2 micro .
- 48 – Create key pair .

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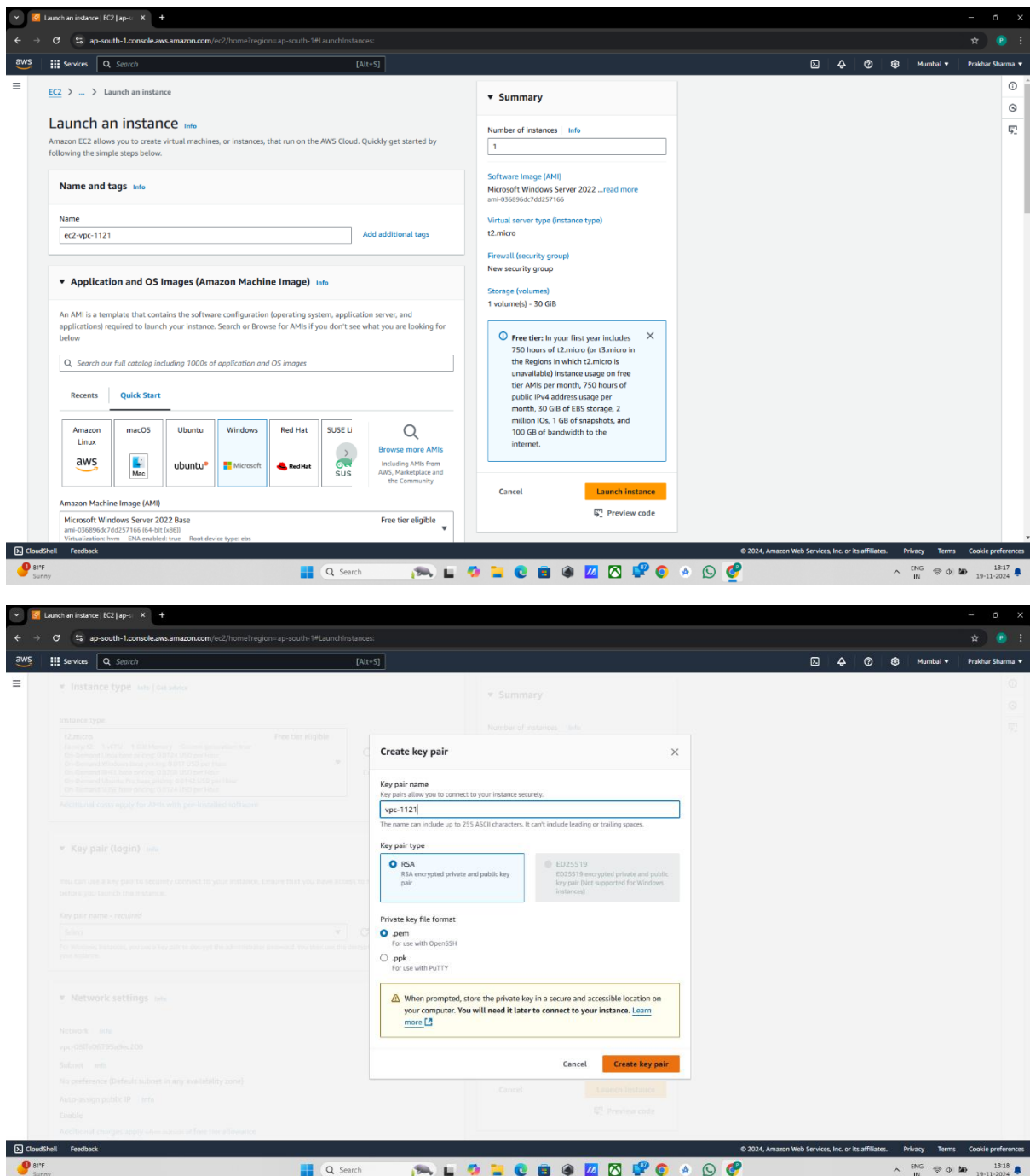
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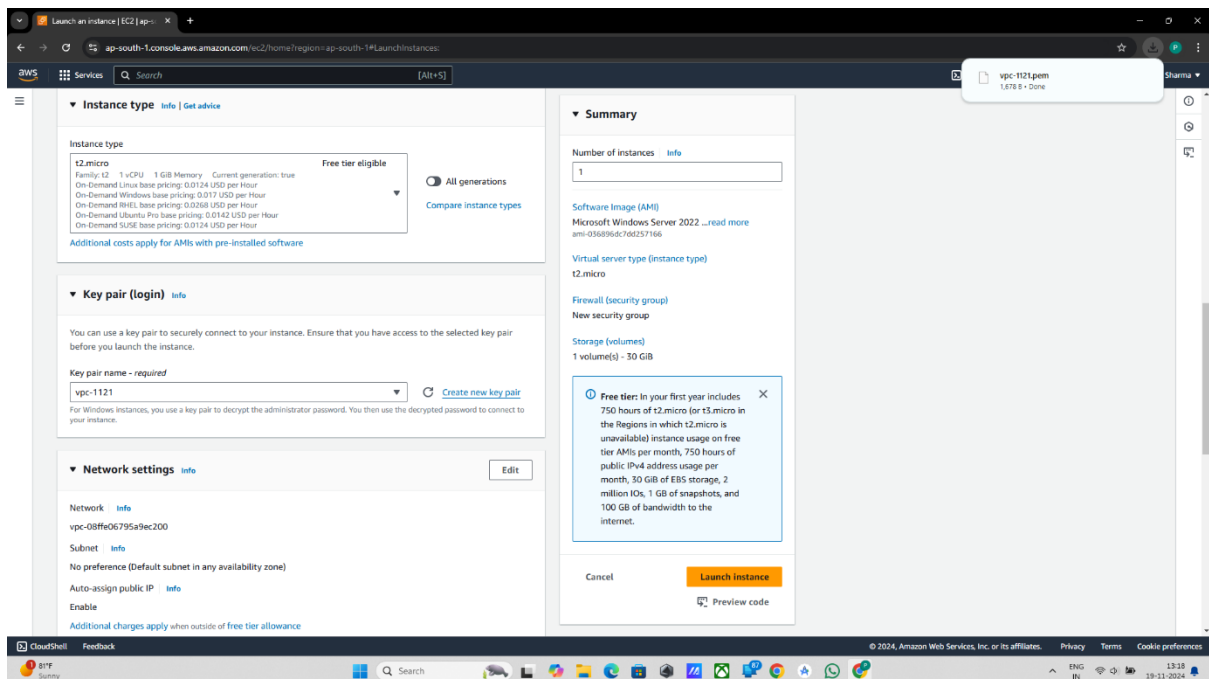
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49 – In Network Settings , click on edit .

50 – Select the VPC created above .

51 – Select the subnet .

52 – Enable auto assign public ip .

53 – Select create Security group .

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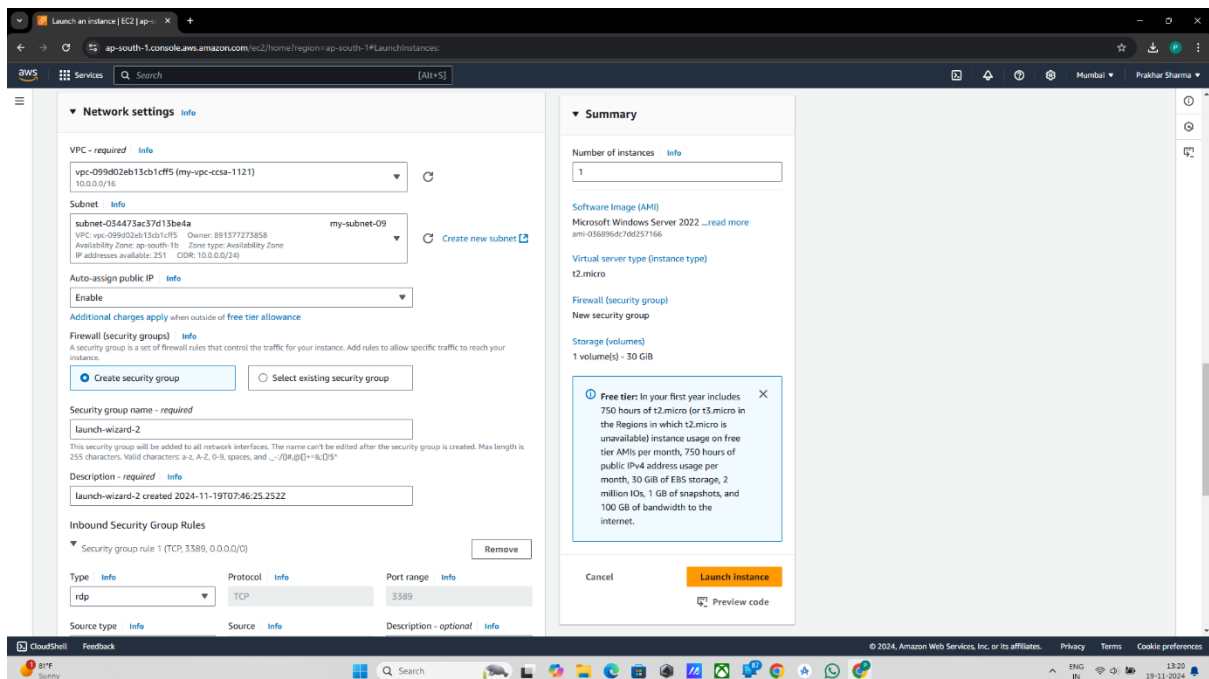
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54 - Add security group rule .

55 – Type : All traffic , Source type : Anywhere .

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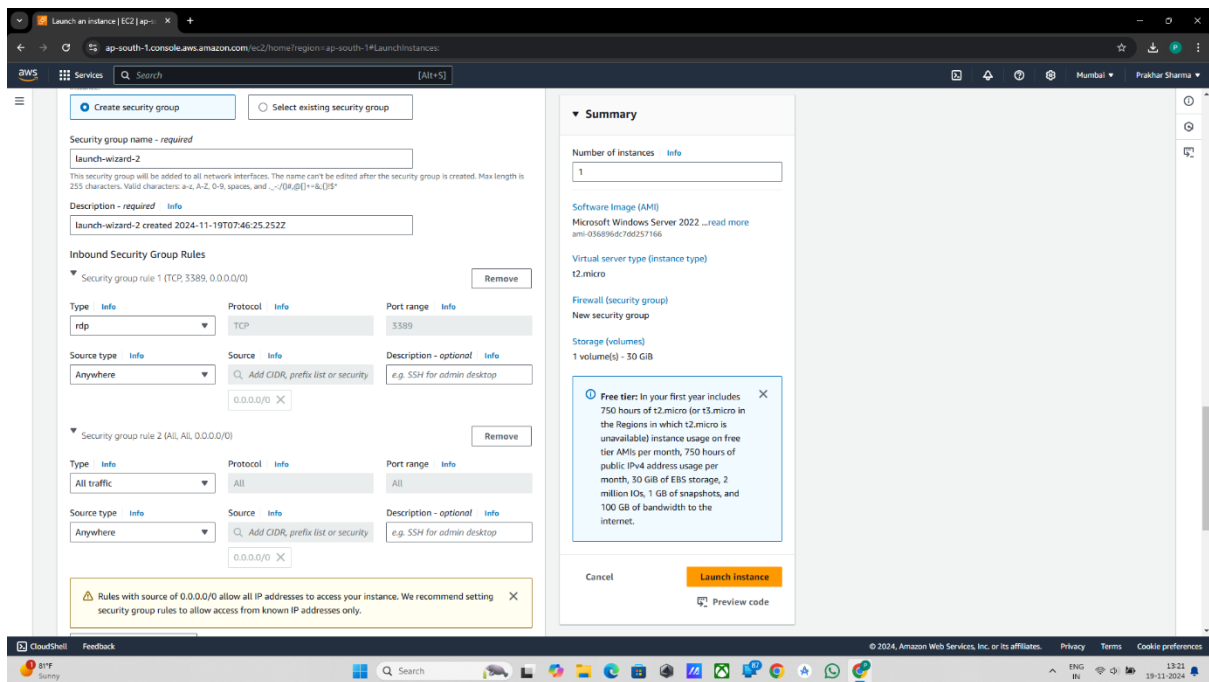
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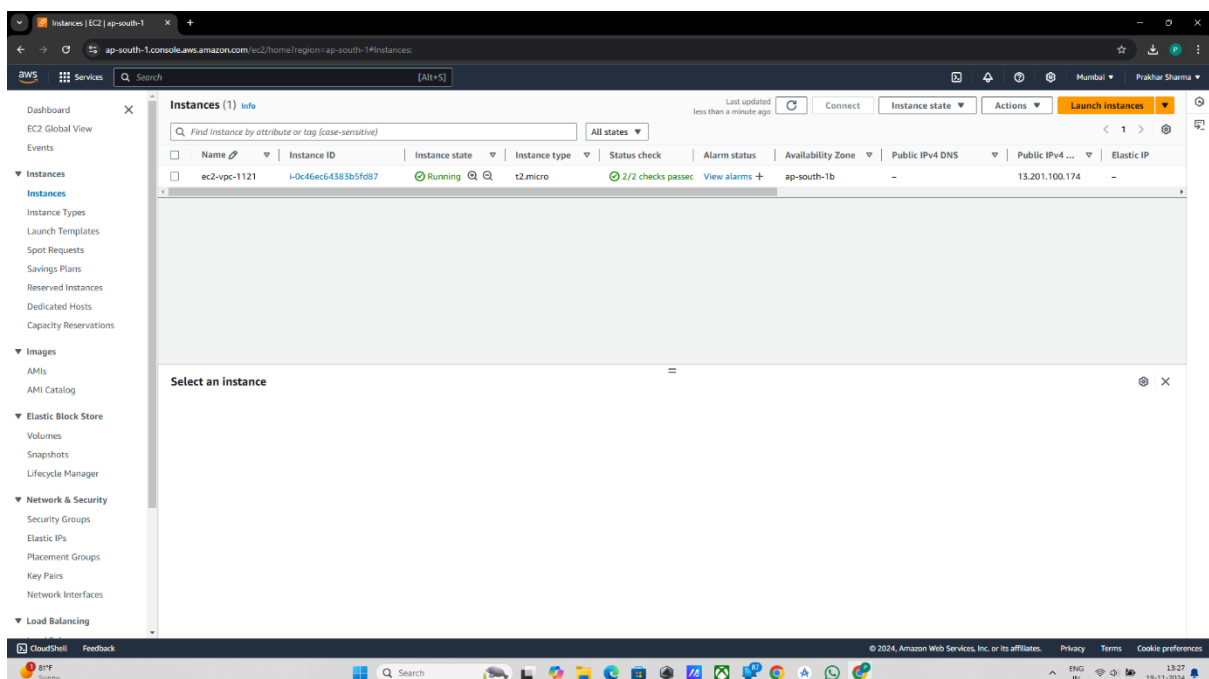
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56 – Click on launch Instance .



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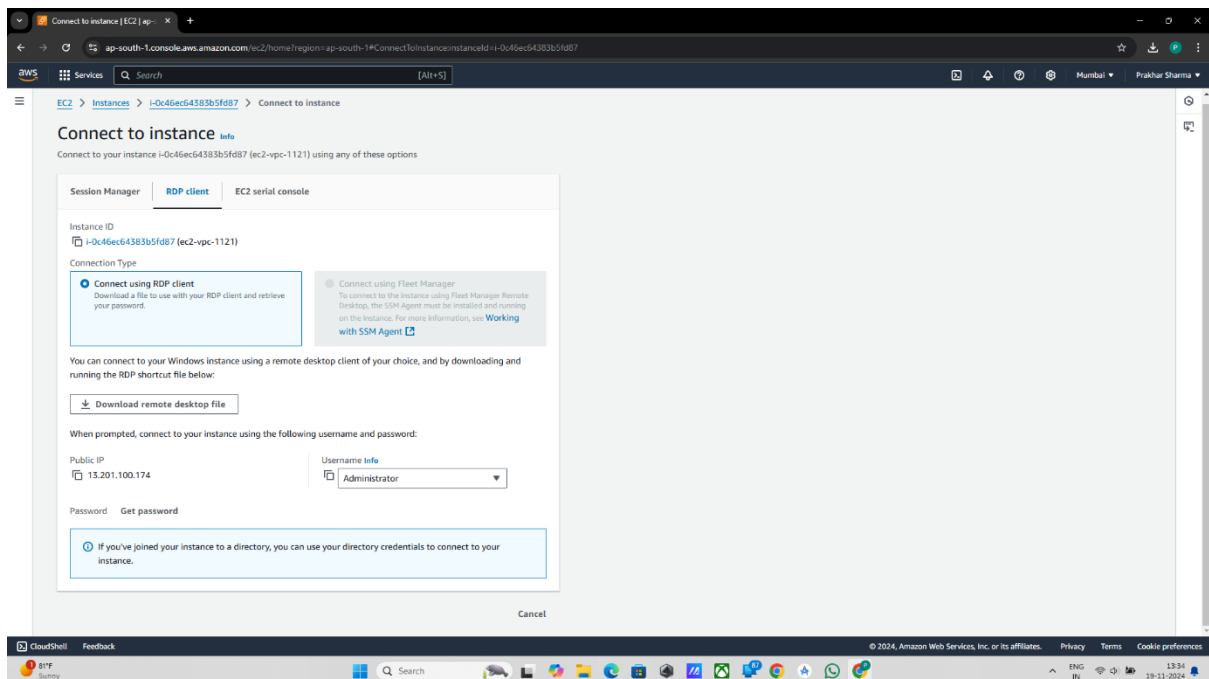
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57 – Select the instance and click on connect .

58 – Download the remote desktop file .

59 – Click on get password .



60 – Upload the private key pair file .

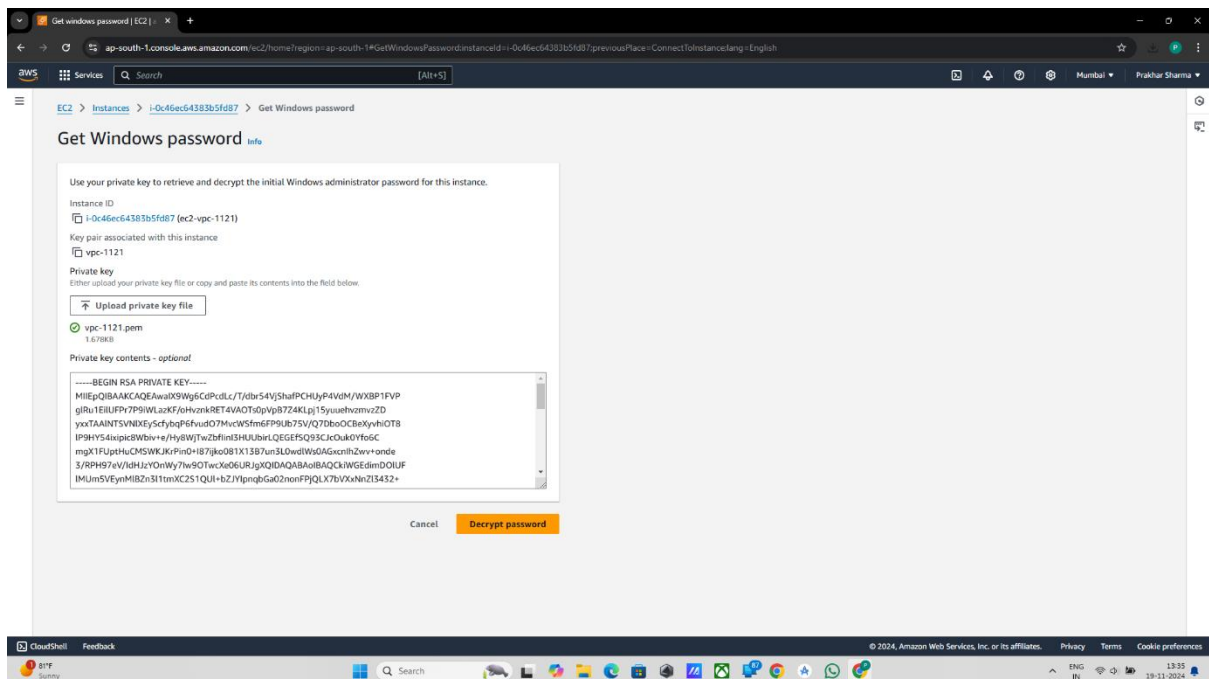
61 – Click on decrypt password .

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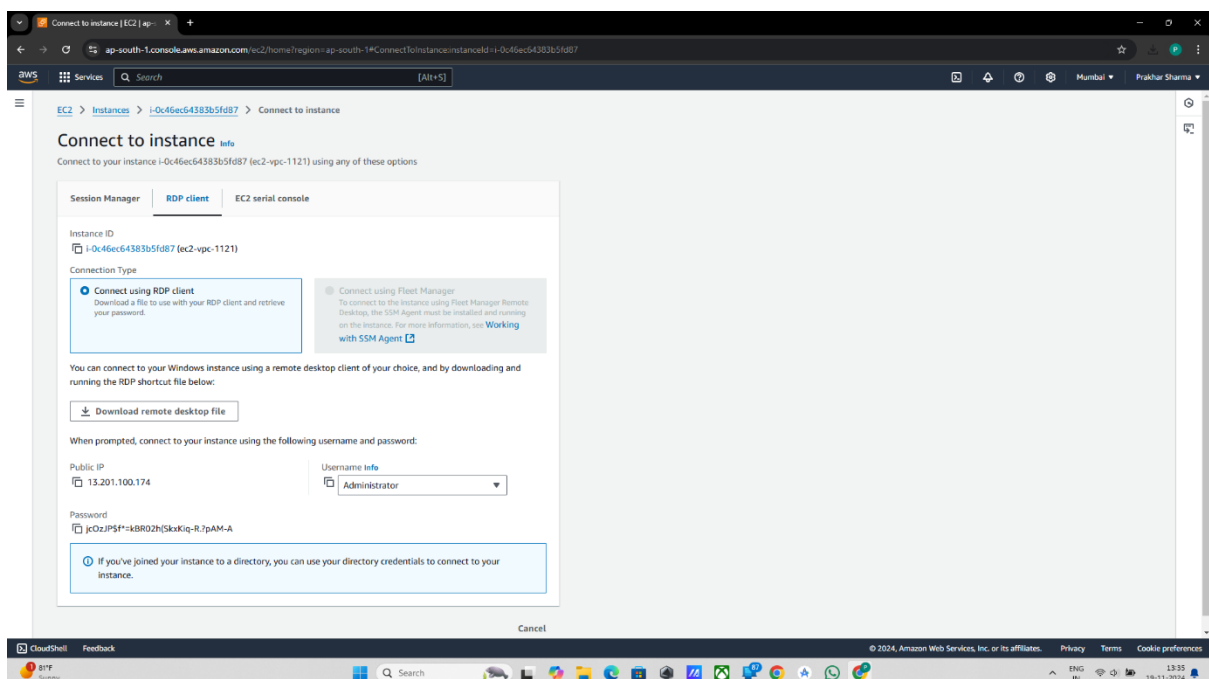
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Copy the password and paste it on notepad .



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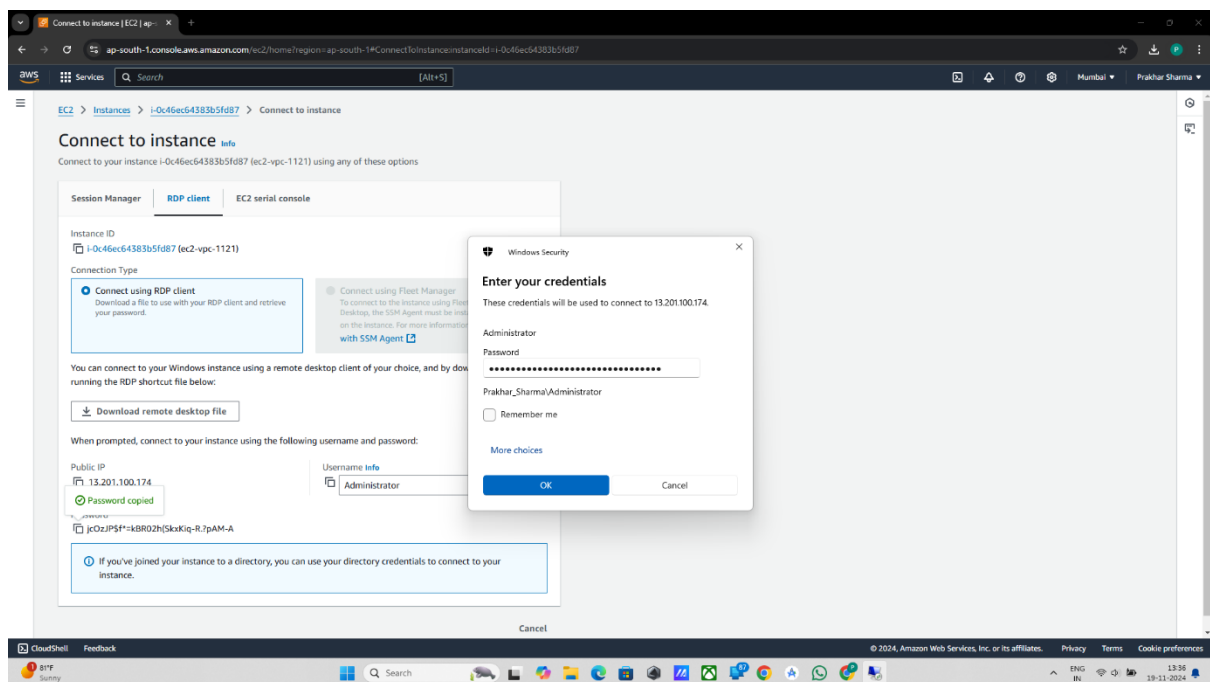
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62 – Open the remote desktop file .

63 – Paste the password .

64 – Click on ok .

65 – Click on Connect .



A new window will open after successful connection .

66 – Now Go to VPC and search Network ACLs in the left hand panel.

67 – Click on it .

68 – Click on Create Network ACLs .

69 – Name the NACL .

70 – Select the VPC from dropdown .

71 – Click on create Network ACL .

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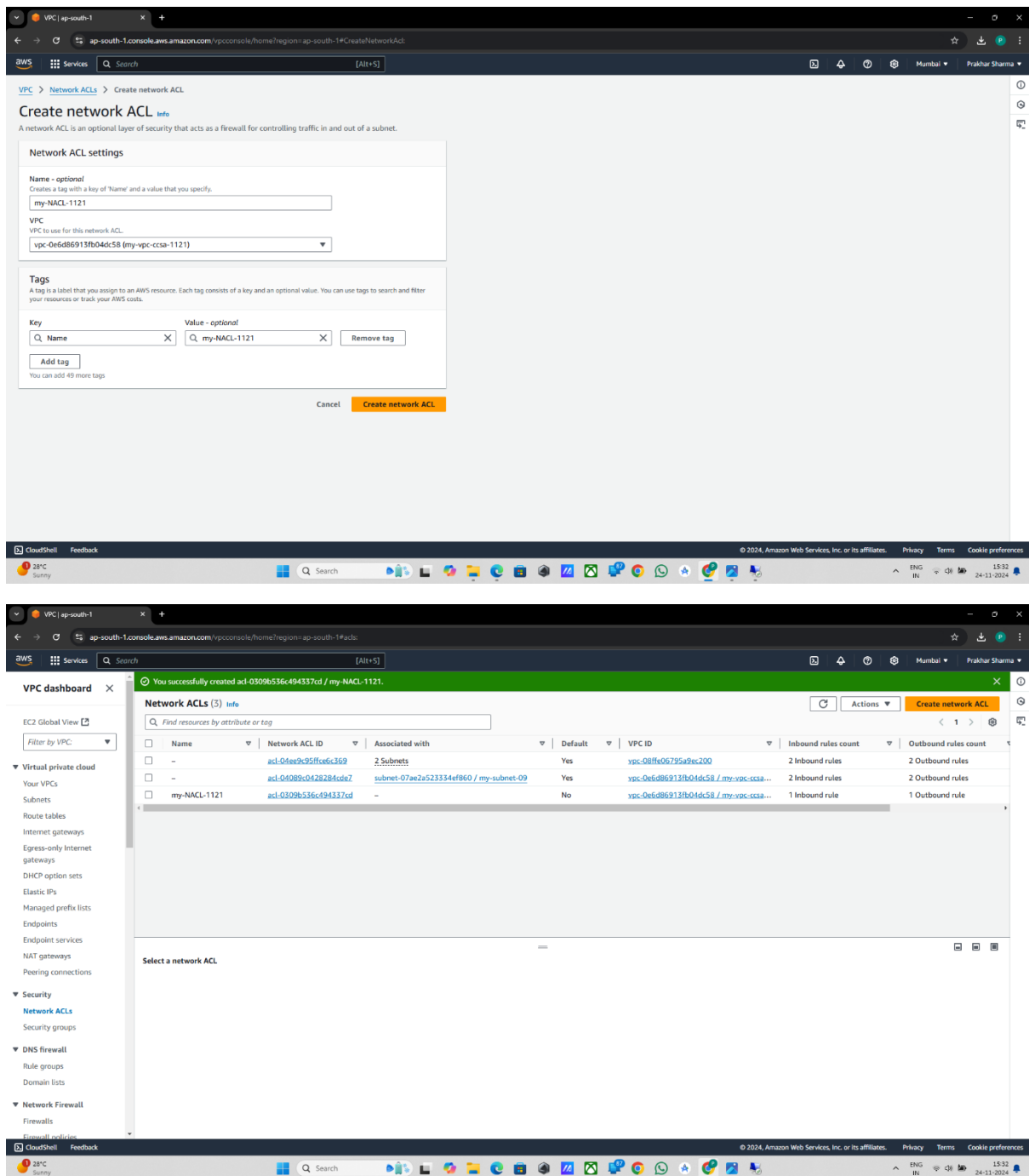
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The screenshot displays the AWS Management Console interface. The top section shows the 'Create network ACL' wizard in the 'ap-south-1' region. The 'Network ACL settings' section includes a 'Name' field with the value 'my-NACL-1121' and a 'VPC' dropdown menu showing 'vpc-0e6d86913fb04dc58 (my-vpc-ccsa-1121)'. The 'Tags' section shows a key-value pair 'Name' with value 'my-NACL-1121'. The bottom section shows the 'Network ACLs' list with a table containing the following data:

Name	Network ACL ID	Associated with	Default	VPC ID	Inbound rules count	Outbound rules count
-	acl-04ee9c35f3ce6c369	2 Subnets	Yes	vpc-08ff606795a9ec200	2 Inbound rules	2 Outbound rules
-	acl-04089c0428284cde7	subnet-07ae2a523334ef860 / my-subnet-09	Yes	vpc-0e6d86913fb04dc58 / my-vpc-ccsa...	2 Inbound rules	2 Outbound rules
my-NACL-1121	acl-0309b536c494337cd	-	No	vpc-0e6d86913fb04dc58 / my-vpc-ccsa...	1 Inbound rule	1 Outbound rule

72 – Select the NACL .

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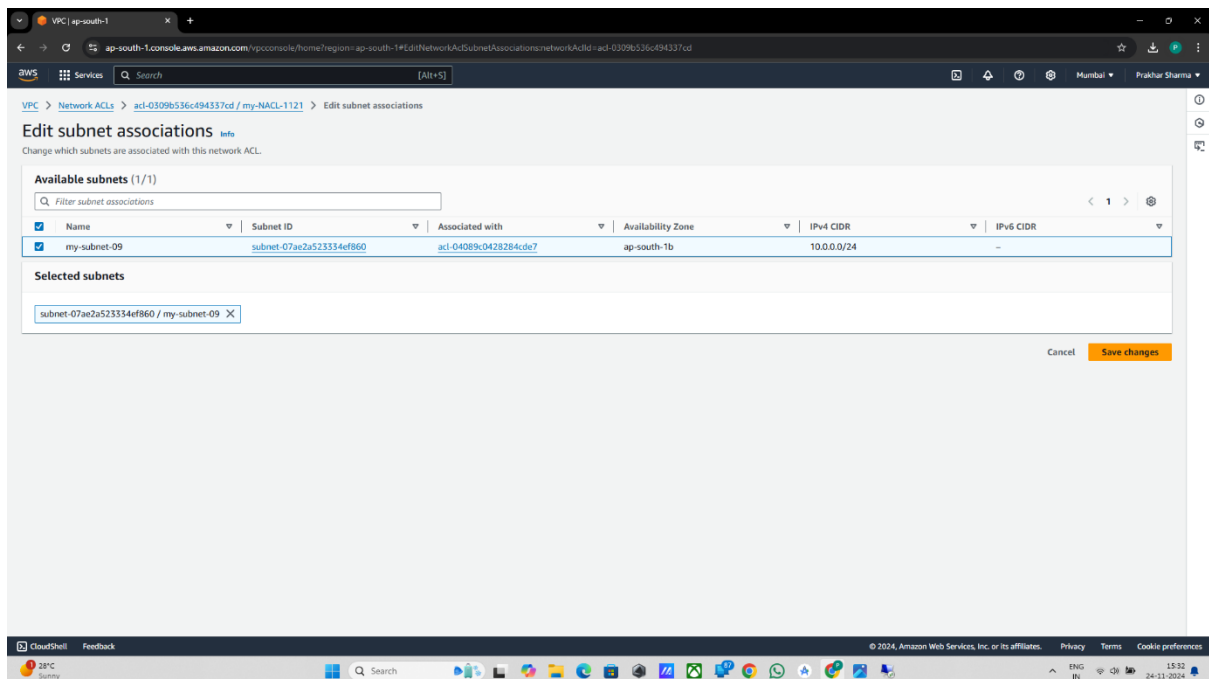
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73 – Go to Subnet Association .

74 – Click on edit association .

75 – Select the subnet and click on save changes .



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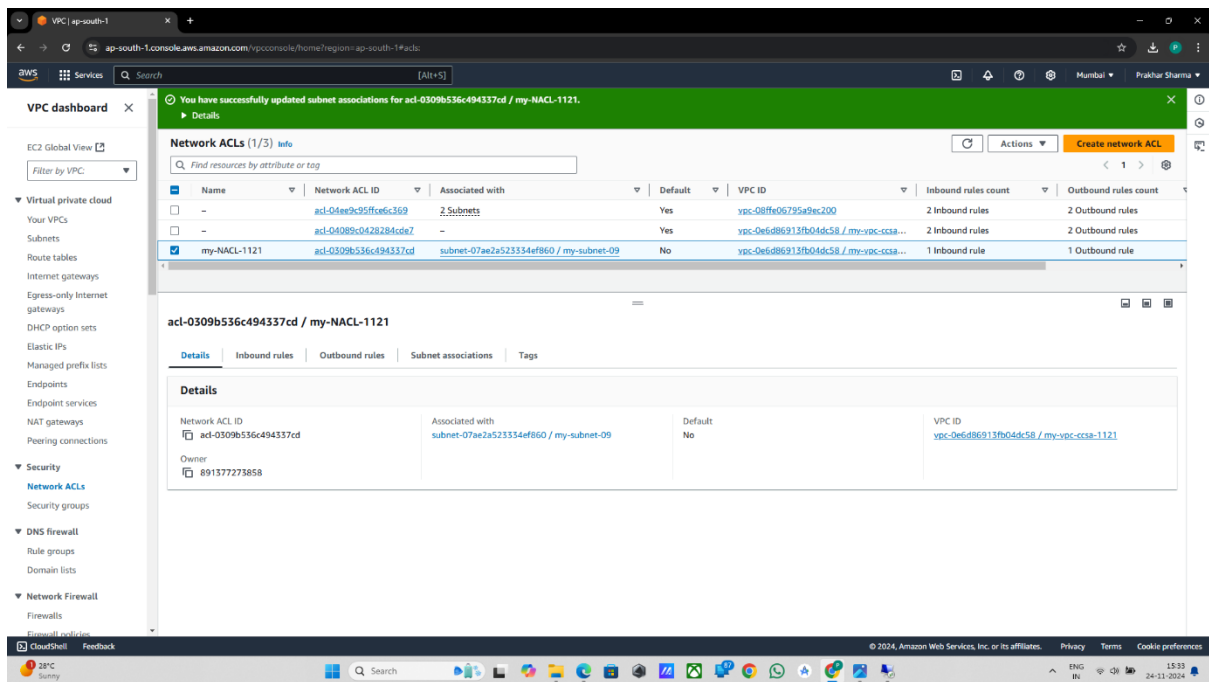
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76 – Go to inbound rule

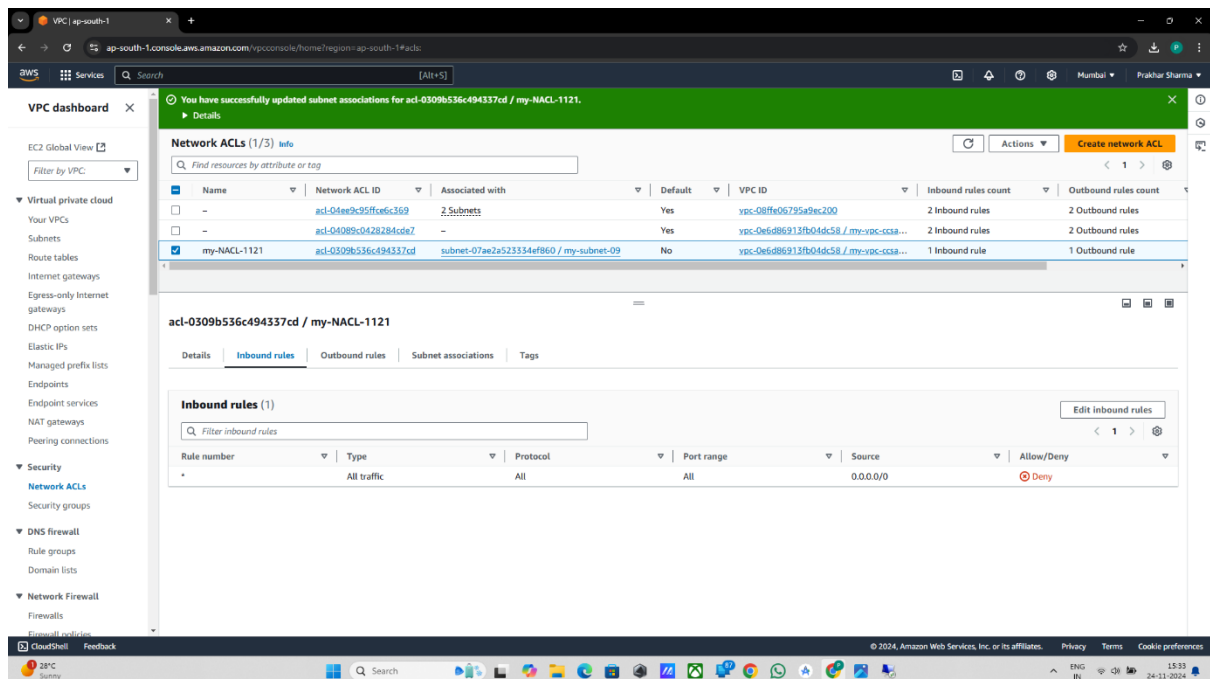
The inbound rule deny the traffic .

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77 – Go to outbound rule .

The outbound rule deny the traffic .



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Network ACLs (1/3) info

Name	Network ACL ID	Associated with	Default	VPC ID	Inbound rules count	Outbound rules count
-	acl-04ee9c95ffce6c369	2 Subnets	Yes	vpc-08ff06795a9ec200	2 Inbound rules	2 Outbound rules
-	acl-04089c0428284cde7	-	Yes	vpc-0e6d86913fb04dc58 / my-vpc-ccsa...	2 Inbound rules	2 Outbound rules
my-NACL-1121	acl-0309b536c494337cd	subnet-07ae2a523334ef860 / my-subnet-09	No	vpc-0e6d86913fb04dc58 / my-vpc-ccsa...	1 Inbound rule	1 Outbound rule

acl-0309b536c494337cd / my-NACL-1121

Outbound rules (1)

Rule number	Type	Protocol	Port range	Destination	Allow/Deny
*	All traffic	All	All	0.0.0.0/0	Deny

We are not able to connect- because we have created custom NACL where everything is denied inbound as well as outbound

Network ACLs (1/3) info

Name	Network ACL ID	Associated with	Default	VPC ID	Inbound rules count	Outbound rules count
-	acl-04ee9c95ffce6c369	2 Subnets	Yes	vpc-08ff06795a9ec200	2 Inbound rules	2 Outbound rules
-	acl-04089c0428284cde7	-	Yes	vpc-0e6d86913fb04dc58 / my-vpc-ccsa...	2 Inbound rules	2 Outbound rules
my-NACL-1121	acl-0309b536c494337cd	subnet-07ae2a523334ef860 / my-subnet-09	No	vpc-0e6d86913fb04dc58 / my-vpc-ccsa...	1 Inbound rule	1 Outbound rule

acl-0309b536c494337cd / my-NACL-1121

Inbound rules (1)

Rule number	Type	Protocol	Port range	Source	Allow/Deny
*	All traffic	All	All	0.0.0.0/0	Deny

Remote Desktop Connection error dialog:

- 1) Remote access to the server is not enabled
- 2) The remote computer is turned off
- 3) The remote computer is not available on the network

Make sure the remote computer is turned on and connected to the network, and that remote access is enabled.

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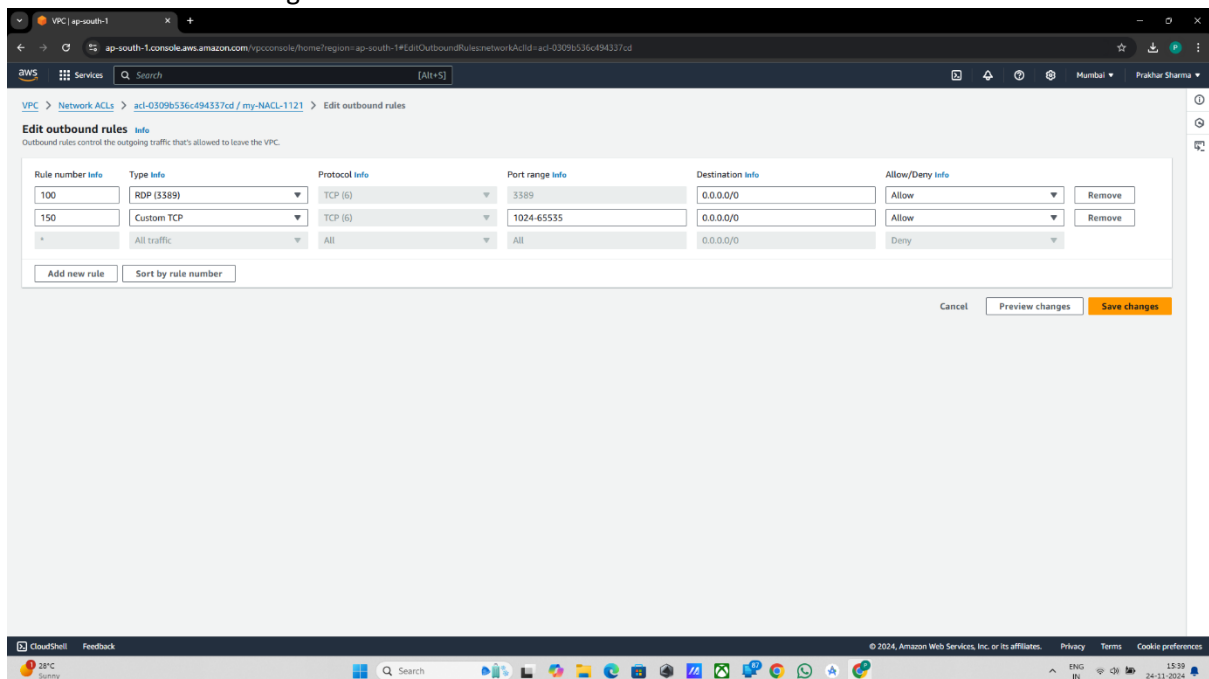
78 - Go to outbound rule and click on edit outbound rules.

79 – Add two rules

- 1st – Rule no.100 , type – RDP , Destination no - 0.0.0.0/0 , Allow/Deny – Allow .

- 2nd – Rule no.150 , type – Custom TCP , Port Range – 1024-65535 , Destination no - 0.0.0.0/0 , Allow/Deny – Allow .

80 – Click on Save Changes .



81 - Go to inbound rule and click on edit inbound rules.

82 – Add two rules .

- 1st – Rule no.100 , type – Custom TCP , Port Range – 1024-65535 , Destination no - 0.0.0.0/0 , Allow/Deny – Allow .

- 2nd– Rule no.10 , type – RDP , Destination no - 0.0.0.0/0 , Allow/Deny – Allow .

83 – Click on Save Changes .

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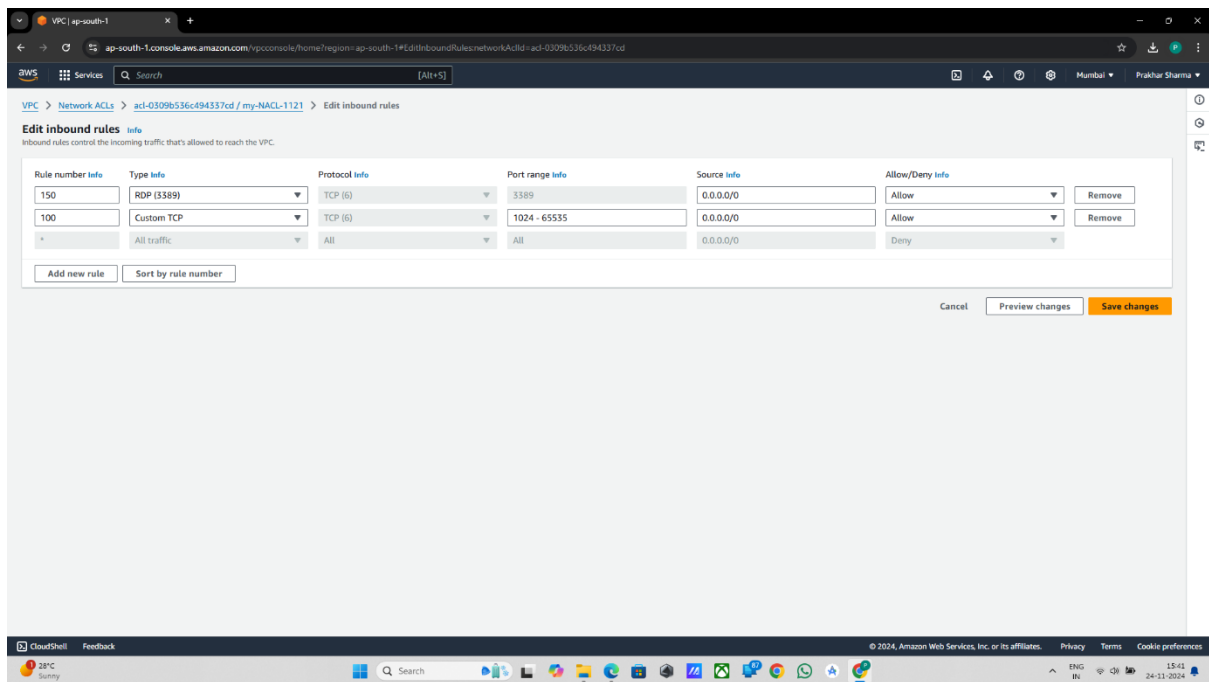
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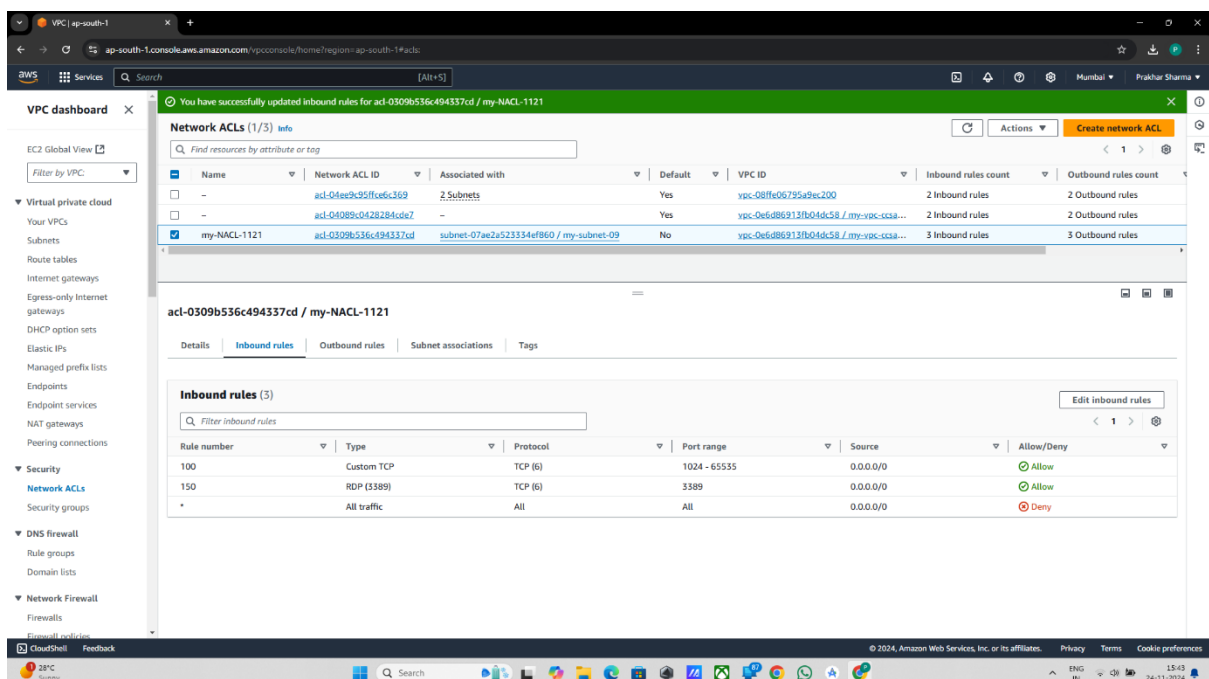
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Inbound rule allowed here



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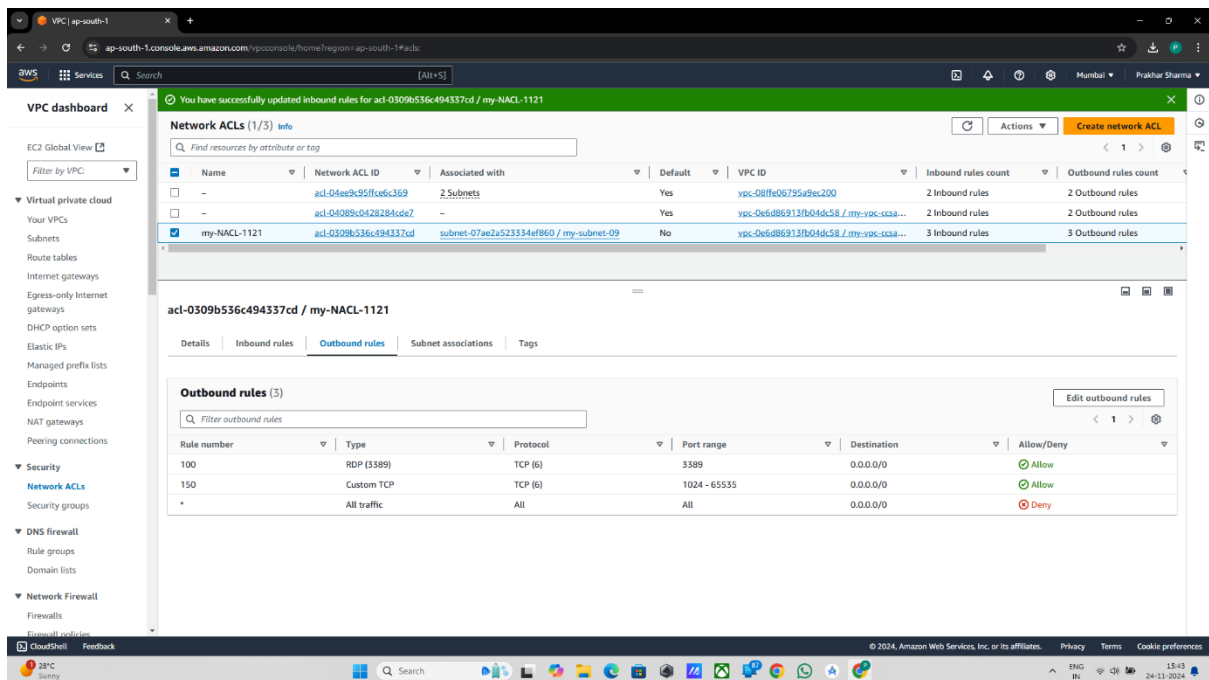
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Outbound rule allowed here



The screenshot displays the AWS Management Console interface for a VPC named 'my-NACL-1121'. The 'Network ACLs' section is active, and the 'Outbound rules' tab is selected. A table lists three outbound rules:

Rule number	Type	Protocol	Port range	Destination	Allow/Deny
100	RDP (3389)	TCP (6)	3389	0.0.0.0/0	Allow
150	Custom TCP	TCP (6)	1024 - 65535	0.0.0.0/0	Allow
*	All traffic	All	All	0.0.0.0/0	Deny

Now go for remote access

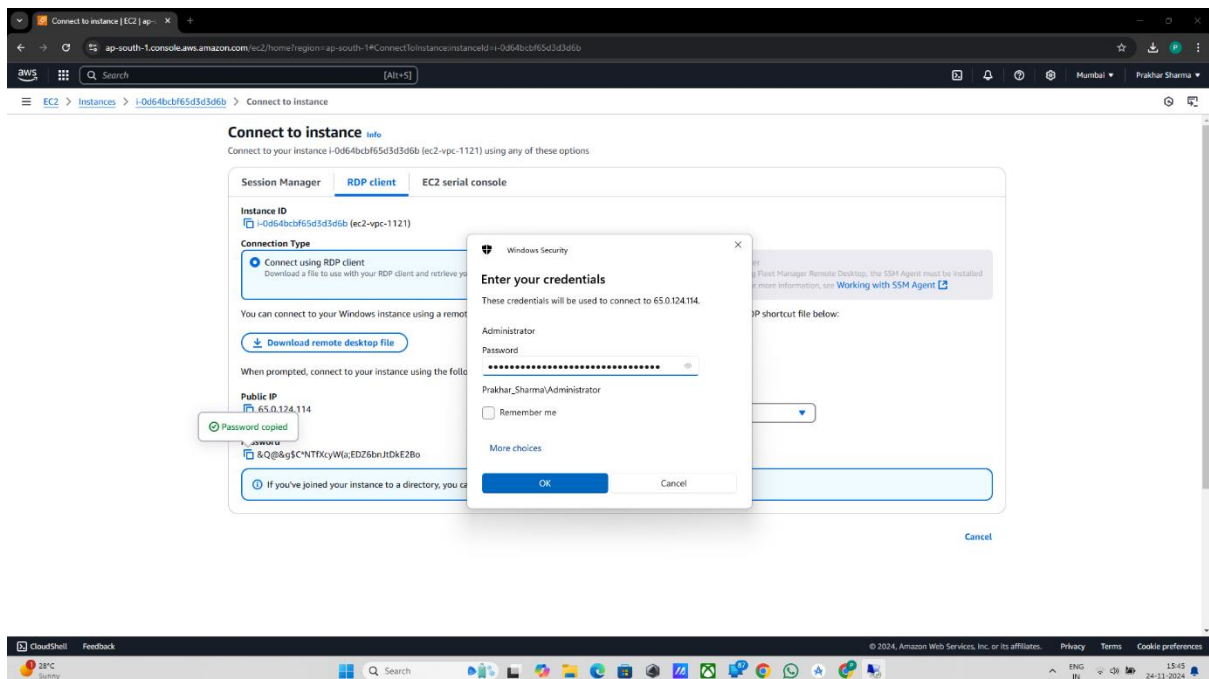
Now you can able to connect

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Now NACL is permitting the RDP also, now we have the double level of security, first we have the NACL permission if we removed it from security group again we are not able to connect RDs to this machine But from both end it is permitted now.so we are able to connect to EC2 instance

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